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RAIL-LESS ATTACHMENT ROOF MOUNT

Abstract

A rail-less roof mounting apparatus comprising a clamp assembly including a first clamp portion configured to clamp against a solar panel module and a second clamp portion configured to engage the first clamp portion. The second clamp portion including a fastener section configured to receive a fastener, a clamp support ledge extending in a first direction, and having a first surface configured to engage with the solar panel module. The clamp support ledge having a soft-clamping arm attached to a second surface of the clamp support ledge, and a protrusion having a surface configured to engage with a bottom surface of a second solar panel module. The protrusion extending from the fastener section in a second direction opposite the first direction and a base that is rotatable with respect to the clamp assembly, wherein the fastener extends through the clamp assembly and the base, the fastener engaging a nut therein.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Application No. 63/562,201, filed Mar. 6, 2024, the entire disclosure of which is incorporated herein by reference and for all purposes.

BACKGROUND

[0002] As the solar energy industry continues to grow, the equipment to mount solar photovoltaic (PV) modules on different types of structures and/or locations continues to adapt and improve as well. Conventional solar PV module mounting assemblies are frequently designed with a specific use according to a particular surface structure. Current mounting assemblies frequently lack adequate wire management, adequate load strength, and the ability to be implemented on multiple different structures and/or locations. As such, conventional mounting assemblies are overly burdensome to install, often requiring excessive amounts of screws and the manual application of sealant. Despite the numerous existing systems for mounting solar panel modules, there is room for improvement.

[0003] Installation of solar PV module mounting assemblies onto a shingled roof presents many challenges. For example, roof shingles are arranged and installed above rafters. Since roof shingles may have various sizes and may be installed above a network of rafters. The various sizes and arrangement of roof shingles over a network of rafters may result in circumstances where shingles may overlap at or near a rafter. Installing a solar PV module mounting assembly on a shingled rooftop that must be mounted to a rafter may be difficult when overlapping shingles create an uneven surface. The uneven surface and proximity of the overlapping shingles relative to the rafters may inhibit the ability for a solar PV module mounting assembly to mount to the rafters.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] FIG. 1 illustrates a top-front perspective view of a solar panel mounting system, including a rail segment and a mount, according to an embodiment of this disclosure.

[0005] FIG. 2A illustrates an isometric side view of a rail-less roof attachment system and a clamp assembly, according to an embodiment of this disclosure.

[0006] FIG. 2B illustrates an isometric front view of the rail-less roof attachment system of FIG. 2A with the base rotated 180° relative to the base depicted in FIG. 2A, according to an embodiment of this disclosure.

[0007] FIG. 3 illustrates an isometric bottom-side view of a rail-less roof attachment system and a clamp assembly with a single-sided upper portion, according to an embodiment of this disclosure.

[0008] FIG. 4 illustrates an isometric bottom-side view of the rail-less roof attachment system and a clamp assembly of FIG. 2A with a double-sided upper portion, according to an embodiment of this disclosure.

[0009] FIG. 5 illustrates a close-up of an isometric bottom-side view of the clamping portion of the clamp assembly of FIG. 2A, according to an embodiment of this disclosure.

[0010] FIG. 6 illustrates an isometric top view of the rail-less roof attachment system and a clamp assembly of FIG. 2A, according to an embodiment of this disclosure.

[0011] FIG. 7 illustrates an isometric back-side view of a rail-less roof attachment apparatus, according to an embodiment of this disclosure.

[0012] FIG. 8 illustrates an isometric front-side view of the rail-less roof attachment apparatus of FIG. 7, according to an embodiment of this disclosure.

[0013] FIG. 9 illustrates an isometric front view of a rail-less roof attachment system and a clamp assembly, according to an embodiment of this disclosure.

DETAILED DESCRIPTION

Overview

[0014] The Detailed Description is set forth with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical items. Furthermore, the drawings may be considered as providing an approximate depiction of the relative sizes of the individual components within individual figures. However, the drawings are not to scale, and the relative sizes of the individual components, both within individual figures and between the different figures, may vary from what is depicted. In particular, some of the figures may depict components as a certain size or shape, while other figures may depict the same components on a larger scale or differently shaped for the sake of clarity.

[0015] This disclosure is directed to a rail-less attachment roof system for mounting solar panel modules “direct-to-deck” (e.g., mounting directly to the roof shingles and sheathing as opposed to the raised and/or exposed rafters). Additionally, this disclosure is directed to a rail-less attachment roof system utilizing solar panel module mounts having a “soft-clamp” feature. For example, before a solar panel module is lifted to a roof, the solar panel module mounts may be arranged on a solar panel module at their installation positions and temporarily secured (e.g., soft-clamped) to the solar panel module, which allows the mounts to remain secured against the solar panel module during transportation. When the solar panel module and the soft-clamped mounts are set in position for installation, base of each solar panel mount may be rotated, as needed, while the clamp assembly portion of each solar panel mount remains soft-clamped to the solar panel module. The system and/or one or more components thereof may alternatively be referred to as an apparatus. Features of the system are further described as shown in the figures and expressed in the claims listing.

[0016] The structures and features described herein address difficulties associated using a direct-to-deck installation method for installing and sealing a roof mount to a roof without installing and sealing over shingle course steps (e.g., localized high points where shingles overlap), securing a mount to the rafters under the sheathing, and minimizing excess roof penetrations. More specifically, the embodiments of mounts disclosed herein may include portions that are rotatable (e.g., 180 degrees, etc.) about a clamp mounting face to avoid sealing over shingle course steps and/or to ensure alignment between the mount and a suitable anchor point (e.g., rafter, truss, etc.). In an embodiment, the rotation may occur about the height adjustment bolt along the center axis of the assembly. The attachment is asymmetric about this axis of rotation in that the distance to mounting holes present on each flange from the axis is different when rotated 180 degrees, allowing one row of mounting holes on either side to mount flat to a shingle course. Upon rotation, the installer may use a second set of installer accessible holes for mounting to the roof. The attachment may also implement a butyl pad to seal the holes drilled, instead of manually applied sealant, thereby minimizing inconveniences of carrying additional materials and extra steps for the installer.

[0017] The embodiments disclosed herein facilitate finding rafters below the decking (e.g., shingles and sheathing) by using a strategic hole pattern within the base of the mount. For example, in an embodiment, a mount may have a 3/4" horizontal spacing and vertical offset between hole pairs to maintain minimum screw distances. Thus, during installation, the installer may align the center hole pair to the pre-drawn rafter line and place the first screw in the outer hole. If this screw does not hit the rafter, the installer may place a second screw in an adjacent hole. When a screw hits, the installer may add the second screw in that successful hole pair to add mounting strength. This

results in a minimum of two screws per attachment and a maximum of four. By attaching the mounts to the rafters, the disclosed system benefits from the additional strength provided by the rafters, which in turn requires fewer mounts. Installing fewer mounts results in fewer roof penetrations. Other direct-to-deck systems, wherein mounts are attached solely to the decking, do not benefit from being attached to rafters and require more mounts, thus producing more roof penetrations. Additionally, in an embodiment, features for integrated wire management may be included.

[0018] Accordingly, one or more advantages of the disclosed mounts may include: elimination of extra sealing means and tilt associated with installing a mount over a shingle course step; reduction of cantilevering of load relative to screw mount points associated with a mount installed across a shingle course step; and reduction of the difficulty of finding rafters to use for installing roof mounts by allowing installers with a pattern resulting in a smaller margin for error when finding rafters, which will speed up installation, reduce roof penetrations, and reduce the amount of time an installer spends in a fall hazard area.

[0019] Notably, concepts for water sealing/waterproofing technologies for use with fasteners and attachments for solar PV systems for pitched-roof applications may also be applied to flat roofs.

[0020] Attachment components for supporting and securing either rail-mounted or rail-less mounted solar PV modules to pitched roof structures require sealing methods to prevent water ingress at the point where the roof covering has been penetrated by fasteners. Roof coverings may include asphalt shingle, tile, membrane, or metal roofs, among others.

[0021] While flashings have traditionally been used, market solutions to waterproof roof covering penetrations have expanded beyond flashing solutions. Such non-flashing solutions may ease installation and increase speed of installation. Non-flashing sealing technologies may include butyl tape, sealant (e.g., one-component polymeric flowable sealant), non-flowable sealant, seals formed by mechanical means, or any other solution that would prevent water intrusion.

[0022] Concepts described herein may improve on existing water sealing/waterproofing technology for roof attachments by leveraging designs that improve one or more of: ease of installation, installation speed, production cost, transportation cost, packaging considerations (e.g., smaller packaging design, less mess, less material for packaging), performance, and/or durability.

Illustrative Embodiments

[0023] FIG. 1 illustrates a top-front perspective view of a solar panel system **100** (“system **100**”), including one or more solar panel module frames(s) **102** (intended to secure a solar panel modules) and one or more mount(s) **104**. In an embodiment, a solar panel module may have a top surface and a bottom surface wherein the top surface is oriented such that it is exposed to light and the bottom surface is oriented such that it faces the mounting surface (e.g., roof surface, ground surface, etc.), likewise with the frames **102** that surround the modules and are secured by the mounts **104**. While FIG. 1 illustrates the system **100** being capable of securing a plurality of solar panel modules, the system **100** may secure only a single solar panel module as well. Regardless, as an example, the system **100** depicted here shows multiple frames **102** (for modules) may be disposed above rows of mounts **104**. While the mounts **104** may be used for mounting solar panel modules, it is understood that there may be additional uses (e.g., different objects to mount) for one or more of the components of the mounts **104**.

[0024] FIG. 2A illustrates an isometric side view of a rail-less roof attachment system **200** (“system **200**”) paired with a solar panel module frame F (e.g., with a rectangular frame side having a return flange on the bottom side), which is shown in a secured position, using dashed thick lines to portray a skeletal outline for convenience in the description of the “soft-clamping” aspect of the disclosure.

[0025] In an embodiment, system **200** may include a base **202** and a clamp assembly **204** connected by a fastener **206** (e.g., bolt, machine screw, carriage bolt, or other suitable fastener). In an embodiment, the base **202** may rotate about the fastener **206** while the clamp assembly **204**

remains stationary.

[0026] In an embodiment, the base **202** may include a horizontal portion **208**, extending in a plane, and having a lower surface configured to abut a mounting surface (not expressly depicted, but implied beneath the horizontal portion **208**). The base **202** may further include an integrally formed vertical portion **210** that protrudes from an upper surface of the horizontal portion **208**. In other words, the vertical portion **210** protrudes in a direction transverse to the plane in which the horizontal portion **208** extends (e.g., a third direction).

[0027] The vertical portion **210** (which may also be referred to hereinafter as a first base portion) of the base **202** may include a cavity **212** (e.g., first cavity, void, hollow, through hole, etc.) defined, at least in part, by first side wall **214**, second side wall **216**, and top wall **218**. The vertical portion **210** may further include a cavity **220** (e.g., second cavity, void, hollow, through hole, etc.) defined, at least in part by, the first side wall **214**, the second side wall **216**, a bottom wall **222**, and a bridge wall **224**, (which separates cavity **212** from cavity **220**). In an embodiment, the cavity **212** may be configured/sized to receive a nut **226** that is accommodated on the fastener **206**. The cavity **212** may also be configured such that the nut **226** does not rotate within the cavity **212**. In an embodiment, the cavity **220** may be configured to accommodate electrical wires (not shown) passing therethrough. In an alternative embodiment (not shown), instead of a nut, the top wall **218** may include a threaded aperture **228** (threading not visible) extending therethrough configured to engage with the fastener **206**.

[0028] In an embodiment, the horizontal portion **208** (which may also be referred to hereinafter as a second base portion) of the base **202** may include a long end **230** (e.g., first end) and a short end **232** (e.g., second end). The long end **230** may include at least one mounting hole (see row of mounting holes **234**). As shown, when there is more than one mounting hole, the plurality of mounting holes may be arranged, for example, in a row of mounting holes **234** (e.g., first row of holes, apertures, divots, openings, marks, etc.). The short end **232** may likewise include one or more mounting holes (see row of mounting holes **236**). Similar to the long end **230**, when there is more than one mounting hole on the short end **232**, the plurality of mounting holes may be arranged, for example, in a row of mounting holes **236** (e.g., second row of holes, apertures, divots, openings, marks, etc.). The mounting holes (in rows **234/236**) extend through the horizontal portion **208** to open adjacent the mounting surface. In an embodiment, a distance **238** (e.g., first length, first distance, etc.) between the row of mounting holes **234** and a central point **242** of the vertical portion **210** may be greater than a distance **240** (e.g., second length, second distance, etc.) between the row of mounting holes **236** and the central point **242** of the vertical portion **210**.

[0029] In an embodiment, the clamp assembly **204** may include an upper clamp portion **244** (which may also be referred to hereinafter as a first clamp portion) and a lower clamp portion **246** (which may also be referred to hereinafter as a second clamp portion). In an embodiment, the upper clamp portion **244** may be configured to engage with a top surface of a solar panel module frame, and the lower clamp portion **246** may be configured to engage with a bottom surface of a solar panel module frame. Thus, the upper clamp portion **244** and the lower clamp portion **246** function together to clamp a solar panel module frame therebetween. In an embodiment, the distance between the upper clamp portion **244** and the lower clamp portion **246** may be adjusted to accommodate the differing heights of a variety of sizes of solar panel modules/frames.

[0030] The upper clamp portion **244** may include a first side wall **248** and a second side wall **229** fixed in parallel by a top wall **252**. A height adjustment assembly **254** extends through an aperture within the top wall **252** of the upper clamp portion **244** (the “aperture” for the height adjustment assembly is not visible as a portion of the height adjustment assembly **254** is shown extending therethrough). The upper clamp portion **244** further includes a first clamp wing **256**, and a second clamp wing **258**, extending respectively from opposite sides of the top wall **252**. The first clamp wing **256** and the second clamp wing **258** are positioned to clamp down on the upper surface of adjacent solar panel frames being mounted therein.

[0031] In an embodiment, the second clamp wing **258** may include a convexity **260** (e.g., point, bulge, etc.) configured to scratch, mar, or otherwise compromise the coating of the solar panel module (e.g., scratch an anodized coating, remove a paint layer, etc.) to improve grounding (e.g., electrical grounding, electrical bonding, etc.). In an embodiment, the convexity **260** may be separate from the second clamp wing **258**. For example, the convexity **260** may be a pin, set screw, or other object protruding from an aperture or cavity within the second clamp wing **258**. In an embodiment not shown, the second clamp wing **258** may include a threaded aperture and the convexity **260** may include a set screw configured to engage with the threaded aperture within the second clamp wing **258**, such that the set screw may extend past the surface of the second clamp wing **258** to scratch, mar, or otherwise compromise the coating of the solar panel module.

[0032] The upper clamp portion **244** may also include aperture **262** disposed in a top surface of the upper clamp portion **244**. Aperture **262** may be sized to allow the threaded fastener **206** to pass through the upper clamp portion **244** into the lower clamp portion **246**. It is also understood that aperture **262** may also be configured and/or sized to allow a tool (e.g., Allen wrench, screwdriver, t-wrench, torx-tip driver, etc.) to pass through in order to access and rotate the head **264** of the threaded fastener **206**.

[0033] In an embodiment, the height adjustment assembly **254** may include a fastener configured to rotate within a threaded aperture such that rotating the fastener clockwise may increase the distance between the upper clamp portion **244** and the lower clamp portion **246** and rotating the fastener counterclockwise may decrease the distance between the upper clamp portion **244** and the lower clamp portion **246**.

[0034] The lower clamp portion **246** may include a fastener section **266**, a soft-clamping arm **268** (e.g., a first flange, ledge, support landing, rest, catch, etc. for securing/supporting a solar panel module frame), and a protrusion **270** (e.g., a second flange, ledge, support landing, rest, catch, etc. for securing/supporting a solar panel module frame on a side opposite the soft-clamping arm **268**). In an embodiment, the soft-clamping arm **268** may extend laterally outward from the fastener section **266** (e.g., in a first direction), and the protrusion **270** may extend laterally outward from the fastener section **266** such that the protrusion **270** extends from a side that is opposite the soft-clamping arm **268** (e.g., in a second direction).

[0035] In an embodiment, the fastener section **266** may be configured to receive the fastener **206** such that the fastener **206** extends vertically through the lower clamp portion **246** via the fastener section **266**. The fastener section **266** may include a first aperture **272** (e.g., top aperture, upper aperture, etc.) and a second aperture **274** (e.g., bottom aperture, lower aperture, etc.), neither of which are not visible in FIG. 2A since the fastener **206** is shown as extending through the first aperture **272** fastener **206** and the second aperture **274**. In an embodiment, the first aperture **272**, the second aperture **274**, or both may be threaded to accommodate the fastener **206**.

[0036] In an embodiment, the soft-clamping arm **268** may include a clamp element **276**. In an embodiment, as discussed in greater detail below, the clamp element **276** may be slidable with respect to an integrally-formed, clamp support ledge **278** (e.g., flange, member, surface, etc.) that protrudes from the fastener section **266**. The clamp support ledge **278** provides a structural attachment location for the clamp element **276**. The soft-clamping arm **268** may further include a return flange **280** (e.g., hook, clasp, etc.) that is configured to extend from the end of the clamp element **276** backwards across the clamp support ledge **278**, and toward the clamp assembly **204**. By so doing, when installed, the return flange **280** is positioned to extend toward the frame edge of a solar panel module, such that the return flange **280** is slidably positionable between two positions. With the return flange **280** in a first position, a solar panel module frame freely passes to fit against the clamp support ledge **278** and laterally within the space between the return flange **280** and the first side wall **248**. With the solar panel module frame in place, the clamp element **276** may be laterally shifted (e.g., pulled, slid, bias released, etc.) into a second position of the return flange **280**, so that the return flange **280** extends over the inside of the frame, and thereby “soft” clamps

the solar panel module to the rail-less roof attachment system **200**. In other words, as indicated above, it is a “soft” clamp because it is secured to the solar panel frame, yet simply and quickly releasable and repositionable—without needing a tool—along the solar panel frame, if needed. [0037] In an embodiment, the clamp support ledge **278** may include a convexity **282** (e.g., point, bulge, etc.) protruding from an upper surface of the clamp support ledge **278** and configured to scratch, mar, or otherwise compromise the coating of the solar panel module (e.g., scratch an anodized coating, remove a paint layer, etc.) to improve grounding (e.g., electrical grounding, electrical bonding, etc.). In an embodiment, the convexity **282** may be separate from the clamp support ledge **278**. For example, the convexity **282** may be a pin, set screw, or other object protruding from an aperture or cavity within the clamp support ledge **278**. In an embodiment, the clamp support ledge **278** may include a threaded aperture and the convexity **282** may include a set screw configured to engage with the threaded aperture within the clamp support ledge **278** such that the set screw may extend past the surface of the clamp support ledge **278** to scratch, mar, or otherwise compromise the coating of the solar panel module.

[0038] In an embodiment, the fastener section **266** of the lower clamp portion **246** may be configured to accommodate the threaded fastener **206**. Moreover, the threaded fastener **206** may be oriented such that a first end of the threaded fastener **206** remains above the lower clamp portion **246**, while a second end of the threaded fastener **206** engages with a nut **226** within the base **202**. In an embodiment, the rotation of the threaded fastener **206** within the nut **226** may cause the clamp assembly **204** to tighten against the base **202**. This rotation is possible while still permitting the soft-clamping arm **268** maintain engagement with the underside frame edge of a solar panel module.

[0039] Similarly, in an embodiment, the base **202** may be rotatable as needed (e.g., 90 degrees, 180 degrees, etc.), again without interference from structure or function of the soft-clamping arm **268**. That is, the base **202** may be rotated **180** degrees while the clamp assembly **204** remains in the same location (e.g., stationary) relative to a central vertical axis within the base **202**. For example, in an embodiment, the clamp assembly may be oriented to attach to a solar panel module wherein the distance from the edge of the solar panel module to the nearest shingle S course step is large. In such cases where the shingle S course step is not near the edge of the solar panel module, it may be advantageous for the base **202** to be oriented such that the long end **230** extends to the shingle course step, as is depicted in FIG. 2A.

[0040] FIG. 2B illustrates an isometric front view of the rail-less roof attachment system of FIG. 2A (“system **200**”) with the base **202** rotated 180 degrees relative to the base **202** depicted in FIG. 2A. For example, in an embodiment, the clamp assembly **204** may be oriented to attach to a solar panel module frame, and the distance from the edge of the solar panel module to the nearest shingle S course step is small. In such cases where the shingle S course step is near the edge of the solar panel module, it may be advantageous for the base **202** to be oriented such that the short end **232** extends to the shingle S course step, as is depicted in FIG. 2A. Accordingly, the long end **230** may partially extend under the solar panel module.

[0041] FIG. 3 illustrates an isometric bottom-side view of the rail-less roof attachment system and clamp assembly **300** with a single-sided upper clamp portion **302**. Moreover, note that the view is from the opposite direction as both of FIGS. 2A and 2B.

[0042] In an embodiment, the single-sided upper portion **302** may include a top **304** and a side wall **306** that integrally intersects the top **304**. The lower clamp portion **246** may be similar or the same as the lower clamp portion **246** discussed with respect to FIGS. 2A and 2B, (including the fastener section **266**, the soft-clamping arm **268**, and the protrusion **270** for example). In more detail, the fastener section **266** may be defined structurally by a side wall **308** (e.g., first side wall, etc.) and a side wall **310** (e.g., second side wall, etc.) opposite the side wall **308**. The outside surface **312** of side wall **310** is configured to engage with an outside surface of a frame of a solar panel module. The clamp support ledge **278** extends from the outside surface **312** of the side wall **310** in a

direction away from the side wall **310**. Additionally, a tab **314** protrudes downward from the clamp support ledge **278**, through a surface (in a slot, see FIG. 5, discussed hereinbelow) of the clamp element **276**, and toward the base **202**.

[0043] In an embodiment, the clamp element **276** may include a body **316**, a retaining rod **318**, and a biasing member **320** (e.g., coil spring, etc.). The body **316** is secured to an underside of the lower clamp portion **246**. In an embodiment, the body **316** may include a first side wall **322**, a second side wall **324** opposite the first side wall **322**, and a rear wall **326** that extends from the first side wall **322** to the second side wall **324**. The retaining rod **318** may be anchored (e.g., screwed in, pressed in, screwed through, etc.) in the rear wall **326** and extend along the length of the body **316** between the first side wall **322** and the second side wall **324** and through an aperture within the tab **314**.

[0044] In an embodiment, the clamp element **276** may be extended and/or retracted slidingly, in a lateral direction to soft-clamp to the solar panel module frame. Thus, the clamp element **276** may be positioned against a side of the solar panel module and the return flange **280** (e.g., hook) may be wrapped (e.g., curve, bent, etc.) around the inside lip/flange (e.g., edge, shelf, ledge, etc.) of the underside or the bottom surface of the solar panel module frame. In an embodiment, the clamp element **276** may slide in the direction **328** or the direction **330**, as required, for the return flange **280** to be engaged with the solar panel module frame. For example, a solar panel module may be positioned within the system **200** while the clamp element **276** is fully extended in the direction **330** such that the side of the solar panel module abuts against the side wall **310** and the clamp support ledge **278**.

[0045] In an embodiment, prior to solar panel module installation, the clamp element **276** may be fully slid in the direction **328**. The clamp element **276** of the soft-clamping arm **268** may be pressed, by hand, in the direction **330** to an adequate width for accommodating a solar panel module, as explained above. As the clamping portion moves in the direction **330**, a spring **320** may compress between the rear wall **326** and the tab **314**, thereby creating a biasing force. When the solar panel module is in a proper position for installation (e.g., against the outside surface **312** and the clamp support ledge **278**) the clamping portion may be released. The biasing force generated by the compression of spring **320** may force the clamp element **276** to slide in the direction **330**, thereby securing (or “soft-clamping”) the solar panel module between the side wall **310** and the clamp element **276**. In an embodiment, when the clamping portion slides in the direction **328** or the direction **330** clamp element **276**, the retaining rod **318**, being anchored to the rear wall **326** of the body **316**, may likewise move with the body **316**. While the retaining rod **318** moves in the direction **328** or **330**, the retaining rod may freely move through the aperture in the tab **314**.

[0046] In an embodiment the spring **320** may have a first end and a second end. In an embodiment, the spring **320** may have a first end that is anchored to the tab **314** and a second end that extends from the rear wall **326** of the clamp element **276** to the tab **314**. In an embodiment, at least a portion of the spring **320** may extend through the tab **314**. In an alternative embodiment, the entire spring **320** may be disposed between the rear wall **326** and the tab **314**.

[0047] FIG. 4 illustrates an isometric bottom-side view **400** of the rail-less roof attachment system of FIG. 2A (“system **200**”) with a double-sided upper portion **402**.

[0048] In an embodiment, the double-sided upper portion **402** may include a top **404**, a side wall **406** (e.g., a first side wall) extending in a direction that is perpendicular to the top **404**, and a side wall **408** (e.g., a second side wall) extending in a direction that is perpendicular to the top **404** and parallel to the side wall **406**. The side wall **408** may include an outside surface **410** that is configured to planarly engage with a side surface of a solar panel module, when installed.

[0049] The operation of the system **200** utilizing the double-sided upper portion **402** is similar to the operation of system **200** utilizing the single-sided upper portion **302**. In an embodiment utilizing the double-sided upper portion **402**, the solar panel module may be positioned such that the side of the solar panel module abuts against the outside surface **410** of the side wall **408**. The

clamp element **276** may be pressed, by hand, in direction **328** towards the side wall **408**, thereby securing (or “soft-clamping”) the solar panel module between the side wall **408** and the clamp element **276**.

[0050] FIG. **5** illustrates a close-up of an isometric bottom-side view of the clamp element **276** of the clamp assembly of FIG. **2A**.

[0051] In an embodiment, the clamp element **276** may include a first fastener **500**, a second fastener **502**, a first fastener slot **504**, a second fastener slot **506**, and a tab slot **508**. The first fastener **500** and the second fastener **502** may be screwed into a bottom surface **510** of the clamp support ledge **278**. The first fastener slot **504** may be sized to accommodate a shaft of the first fastener **500**. The second fastener slot **506** may be sized to accommodate the shaft of the second fastener **502**. The tab slot **508** may be sized to accommodate the tab **314**. In an embodiment, the tab **314** may extend from the bottom surface **510** of the clamp support ledge **278** through the tab slot **508**.

[0052] In an embodiment, the first fastener slot **504**, the second fastener slot **506**, and the tab slot **508** may extend along the length of the body **316**. In an embodiment, when the clamping portion is fully inward (e.g., the spring **320** is fully extended) the first fastener **500**, the second fastener **502**, and the tab **314** may be at a first end of the first fastener slot **504**, second fastener slot **506**, and tab slot **508**, respectively. When the clamp element **276** is fully outward (e.g., the spring **320** is fully compressed), the first fastener **500**, the second fastener **502**, and the tab **314** may be at a second end of the first fastener slot **504**, second fastener slot **506**, and tab slot **508**, respectively.

[0053] FIG. **6** illustrates an isometric top view of the rail-less roof attachment system of FIG. **2A** (“system **200**”). In an embodiment, the row of mounting holes **236** may include one or more individual mounting holes (**236a**, **236b**, **236c**, **236d**, **236e**). In an embodiment, the row of mounting holes **236** may include a 0.75” hole spacing hole pattern between five mounting holes (e.g., individual mounting holes **236a**, **236b**, **236c**, **236d**, **236e**) to allow for some error (e.g., ± 1.5 ”) in the drawn rafter line, which may guarantee that at least one of the individual mounting holes (**236a**, **236b**, **236c**, **236d**, **236e**) may line up with a rafter within a total 3” range. By ensuring that at least one of the individual mounting holes (**236a**, **236b**, **236c**, **236d**, **236e**) lines up with a rafter, the base **202** may be securely fastened to a rafter. It is understood that the row of mounting holes **234** may also include a 0.75” hole spacing hole pattern between five mounting holes (e.g., individual mounting holes **234a**, **234b**, **234c**, **234d**, **234e**).

[0054] FIG. **7** illustrates an isometric back-side view of an embodiment of a rail-less roof attachment apparatus **700** (“mount **700**”). In an embodiment, the mount **700** includes a body **702** which includes a horizontal portion **704** and a vertical portion **706**. The mount **700** may be made from resin, plastic, aluminum, or any other suitable material.

[0055] The horizontal portion **704** may include a primary screw hole pattern **800** (not shown in FIG. **7**, see FIG. **8**) on a first side, wherein the holes extend transversely to the horizontal portion **704** and include hole guides **802** that extend above the planar surface of the horizontal portion **704**, and a secondary screw hole pattern **705** on a second side, wherein the holes are directed perpendicularly through the horizontal portion **704**. The secondary screw hole pattern **705** may be utilized, for example, when the attachment is rotated 180 degrees. Both hole patterns may have, for example, in an embodiment, a 0.75” horizontal spacing between each set of two holes, for easier rafter finding as previously discussed.

[0056] The vertical portion **706** includes a lower part **708** that extends transversely from the horizontal portion **704** towards an edge **710** of the body **702**. The pitch of the lower part **708** may be the same or different than a pitch of the transversely extending primary screw hole pattern **800** with respect to the horizontal portion **704**. The vertical portion **706** further includes an upper part **712** that extends transversely upward from the lower part **708**. The upper part **712** serves as a post-row height adjustment area. That is, the upper part **712**, having a first side (e.g., inside surface) **714** and a second side (e.g., outside surface) **616**, may include a slot **718** which extends through a

thickness direction (e.g., through first side **714** and second side **716**) and longitudinally in a vertical direction to accommodate vertical adjustment of the elements to be attached to the mount **700**. Moreover, the first side **714** and the second side **716** may include a textured surface (e.g., teeth, grooves, etc.) to assist in gripping to attached elements. The slot **718** may further allow for the mount **700** to slide off the clamp, rotate, and reattach in one step in order to avoid shingle course steps.

[0057] The body **702** of the mount **700** may further support the sidewalls **720**, **722**, which flank the vertical portion **706** on opposite sides of the body **702**, extending from the horizontal portion **704** on both the lower part **708** and the upper part **712**. Thus, the support sidewalls may provide lateral and vertical strength to the mount **700**.

[0058] FIG. **8** illustrates an isometric front-side view of the rail-less roof attachment apparatus **700** (“mount **700**”) of FIG. **7**. In an embodiment, as indicated above, the secondary hole pattern **800** may include the hole guides **802** that may extend like a tubular sleeve above the planar surface of the horizontal portion **704**. In an embodiment, the hole guides **802** may be positioned within a range of 45-75 degrees, for example, at an angle of 60 degrees relative to the horizontal portion **704**. Such angled positioning may allow for an installer to access and/or install screws underneath the mount, penetrate the roof deck, and ensure that the screws are torqued after penetration, without over angling the holes such that the component slides when screws rotate to initiate a hole.

[0059] In an embodiment, the primary hole pattern **705** and the secondary hole pattern **800** may include one or more holes, respectively, that are staggered spatially in a horizontal plane and/or angled at a desired pitch.

[0060] FIG. **9** illustrates an isometric front view of a rail-less roof attachment system and a clamp assembly. As shown in FIG. **9**, a clamp assembly **900** may be an element that attaches to the mount **700**. Clamp assembly **900** may include a lower clamp member **902** and a corresponding upper clamp member **904**.

[0061] Lower clamp member **902** may include fastener **905** and an aperture configured to receive fastener **905** (the “aperture” is not visible as the fastener **905** is shown extending therethrough). The fastener **905** and aperture may be positioned within the lower clamp member **902** such that the aperture aligns with the slot **718**, such that the fastener **905**, when extending through the aperture, will extend through the slot **718**. In an embodiment, the aperture within the lower clamp member **902** may be configured to engage with the fastener **905** (e.g., the aperture may be threaded). The fastener **905** and the aperture within the lower clamp member **902** may be configured such that, via rotational movement of the fastener **905** within the lower clamp member **902**, the lower clamp member **902** is tightened against the vertical portion **706** of the mount **700**.

[0062] Upper clamp member **904** may include one or more side surfaces **906**, a top surface **908**, a fastener **910** and a spring **912**. In an embodiment, side surface **906** may be configured to planarly engage with a side of a solar panel module. The top surface **908** may be configured to engage with a top surface of a solar panel module. In an embodiment, fastener **910** may be disposed within an aperture (the “aperture” is not visible as the fastener **910** is shown extending therethrough) disposed in a top surface of the upper clamp member **904** such that, via rotational movement of the fastener **910** within the upper clamp member **904**, the upper clamp member **904** is raised relative to the lower clamp member **902** in order to accommodate a height of a solar panel module. In an embodiment, the aperture within the top surface of the upper clamp member **904** may be configured to engage with the fastener **910** (e.g., the aperture may be threaded). In an embodiment, the spring **912** may be used to provide a soft clamping force to clamp assembly **900** to hold the clamp assembly **900** in place while allowing the installer to slide the clamp assembly **900** along the module edge to align the mount **700** to the rafters.

[0063] In an embodiment, a butyl pad **914** with a pull tab may be included to seal the mount **700** to the mounting surface. The pull tab on the butyl pad **914** may allow for quick and easy removal of a release liner during installation. In an embodiment, one or more wire management channels **916**

may attach (e.g., snap, affix, etc.) to either side of the clamp assembly **900** (e.g., on flanges of the lower clamp member **902**).

Conclusion

[0064] Although several embodiments have been described in language specific to structural features and/or methodological acts, it is to be understood that the claims are not necessarily limited to the specific features or acts described. Rather, the specific features and acts are disclosed as illustrative forms of implementing the claimed subject matter.

Claims

1. A rail-less roof mounting apparatus comprising: a clamp assembly including: a first clamp portion configured to clamp against a top surface of a solar panel module, and a second clamp portion configured to engage the first clamp portion, the second clamp portion including: a fastener section configured to receive a fastener, a clamp support ledge extending in a first direction, having a first surface configured to planarly engage with a bottom surface of the solar panel module, the clamp support ledge having: a soft-clamping arm attached to a second surface of the clamp support ledge, and a protrusion having a surface configured to planarly engage with a bottom surface of a second solar panel module, the protrusion extending from the fastener section in a second direction opposite the first direction; and a base that is rotatable with respect to the clamp assembly, wherein the fastener extends through the clamp assembly and the base, the fastener engaging a nut therein.
2. The rail-less roof mounting apparatus of claim 1, wherein the soft-clamping arm includes: a body, including: a first side wall, a second side wall opposite the first side wall, and a rear wall extending between the first side wall and the second side wall, a retaining rod anchored to the rear wall of the body, a tab extending from the second surface of the clamp support ledge in a fifth direction away from the clamp support ledge, the tab having an aperture configured to accommodate the retaining rod, and a biasing member extending between the rear wall of the body and the tab.
3. The rail-less roof mounting apparatus of claim 1, wherein the base includes: a first end includes a first row of first mounting holes, and a second end includes a second row of second mounting holes.
4. The rail-less roof mounting apparatus of claim 3, wherein: the first row of first mounting holes includes a first individual mounting hole spaced approximately 3/4" from a second individual mounting hole, and the second row of second mounting holes includes a first individual mounting hole spaced approximately 3/4" from a second individual mounting hole.
5. The rail-less roof mounting apparatus of claim 1, wherein the base includes a first base portion and a second base portion, the first base portion including: a first side wall extending in a third direction away from the second base portion, a second side wall opposite the first side wall extending in the third direction, a top wall extending from a top edge of the first side wall to a top edge of the second side wall, and a bridge wall extending from an inside surface of the first side wall to an inside surface of the second side wall, wherein the nut is disposed in a cavity defined by the first side wall, the second side wall, the top wall, and the bridge wall.
6. The rail-less roof mounting apparatus of claim 1, wherein the base includes: a first base portion having a cavity accommodating a nut therein, and a second base portion including: a first end extending from the first base portion in a first direction, the first end having a first length, and the first end having a first mounting hole, and a second end extending from the first base portion in a second direction opposite to the first direction, the second end having a second length that is shorter than the first length, and the second end having a second mounting hole.
7. The rail-less roof mounting apparatus of claim 1, the first clamp portion of the clamp assembly including: a first side wall having a first end and a second end opposite the first end, a first clamp wing extending outwardly from the first end of the first side wall, a second side wall having a first

end and a second end opposite the first end, the second side wall being opposite the first side wall, a second clamp wing extending outwardly from the first end of the second side wall, and a top wall extending from the first end of the first side wall to the first end of the second side wall, the top wall including an aperture configured to accommodate the fastener.

8. A rail-less roof mounting system, comprising: a clamp assembly including: a first clamp portion, a second clamp portion configured to receive the first clamp portion, the second clamp portion including: a clamp support ledge having a first surface configured to engage with a bottom surface of a solar panel module, a soft-clamping arm attached to a second surface of the clamp support ledge, and a return flange extending from an end of the soft-clamping arm in a direction toward the first clamp portion, and a height adjustment assembly configured to adjust a distance between the first clamp portion and the second clamp portion; a base that is rotatable relative to the clamp assembly; and a fastener extending through the clamp assembly and the base, the fastener engaging a nut therein.

9. The rail-less roof mounting system of claim 8, wherein the soft-clamping arm includes: a body, including: a first side wall, a second side wall opposite the first side wall, and a rear wall extending between the first side wall and the second side wall, a tab extending from the second surface of the clamp support ledge in a second direction away from the clamp support ledge, the tab having an aperture configured to accommodate a retaining rod, and a biasing member extending between the rear wall of the body and the tab.

10. The rail-less roof mounting system of claim 8, the height adjustment assembly including: a threaded aperture passing through a first wall of the first clamp portion, and a second fastener having a first end and a second end, the second end passing through the threaded aperture such that: the first end of the second fastener engages the first clamp portion, and the second end of the second fastener engages the second clamp portion.

11. The rail-less roof mounting system of claim 10, wherein rotation of the second fastener relative to the first clamp portion increases or decreases a distance between the first clamp portion and the second clamp portion.

12. The rail-less roof mounting system of claim 8, wherein the base is rotatable around the fastener such that the base may rotate while the clamp assembly remains stationary.

13. The rail-less roof mounting system of claim 8, the soft-clamping arm further including a return flange configured to curve around a ledge of the solar panel module.

14. The rail-less roof mounting system of claim 8, wherein: the first clamp portion includes: a side wall configured to engage with a lateral edge of a first solar panel module, a first clamp wing extending from the side wall outwardly from the side wall, the first clamp wing configured to engage with a top surface of the first solar panel module, and a second clamp wing configured to engage with a top surface of a second solar panel module, the second clamp wing extending outwardly from the first clamp wing, and a side wall of the second clamp portion is configured to engage with a lateral edge of the second solar panel module.

15. A rail-less roof mounting system comprising: a rail-less roof mounting apparatus, including: a clamp assembly, including: a clamp support ledge having a first surface configured to engage with a bottom surface of a solar panel module, and a soft-clamping arm attached to a second surface of the clamp support ledge, the soft-clamping arm including: a first side wall, a second side wall opposite the first side wall, a rear wall extending between the first side wall and the second side wall, a tab extending from the second surface of the clamp support ledge in a direction away from the clamp support ledge, the tab having an aperture configured to accommodate a retaining rod, and a biasing member extending between the rear wall and the tab; a base that is rotatable with regard to the clamp assembly, and a fastener extending through the clamp assembly and the base, the fastener engaging a nut therein.

16. The rail-less roof mounting system of claim 15, wherein the first surface of the clamp support ledge includes a point configured to electrically bond the solar panel module to the clamp

assembly.

17. The rail-less roof mounting system of claim 16, wherein the clamp assembly includes: a fastener section configured to accommodate the fastener therethrough, the fastener section including: a first side wall extending in a third direction, a second side wall extending in the third direction, a top wall having a first aperture therethrough, the top wall extending from a first end of the first side wall to a first end of the second side wall in a fourth direction transverse to the third direction, and a fourth wall having a second aperture therethrough, the fourth wall extending from a second end of the first side wall to a second end of the second side wall in the fourth direction.

18. The rail-less roof mounting system of claim 15, wherein the base includes a first base portion and a second base portion, the first base portion including: a first cavity defined by: a first side wall extending in a third direction away from the second base portion, a second side wall opposite the first side wall extending in the third direction, a top wall having an aperture therethrough, the top wall extending from a first end of the first side wall to a first end of the second side wall, and a bridge wall extending from an inside surface of the first side wall to an inside surface of the second side wall, wherein the nut is disposed in the first cavity, and a second cavity defined by: the first side wall, the second side wall, the bridge wall, and a bottom wall extending from a second end of the first side wall opposite the first end of the first side wall in the third direction to a second end of the second side wall opposite the first end of the second side wall.

19. The rail-less roof mounting system of claim 18, wherein the second cavity is configured to accommodate electrical wiring.

20. The rail-less roof mounting system of claim 15, wherein the clamp assembly further includes a first clamp portion and a second clamp portion, the second clamp portion including: the clamp support ledge, and the soft-clamping arm.
